Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	ATIALUK, Joseph
Date of birth:	12/24/1987
Subject:	020-AGK
Result:	42,5 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2719	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	3651	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
3	2781	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	3628	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
5	3654	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
6	1577	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
7	597	21.2.3.1	Design and construction	1,00	1,00
8	2422	21.2.3.3	Loads, stresses flutter	1,00	1,00
9	14	21.3.2	Hydraulic systems	0,00	1,00
10	900	21.3.2.2	System components: etc	0,00	1,00
11	1323	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
12	25	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
13	836	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
14	854	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
15	2929	21.8	FUEL SYSTEM	0,00	1,00
16	1408	21.8.2.2	Design, operation, system components, etc	1,00	1,00
17	926	21.9.1.6	Electromagnetism	0,00	1,00
18	817	21.9.3.1	DC Generation	1,00	1,00
19	883	21.13	OXYGEN SYSTEMS	1,00	1,00
20	1601	21.13	OXYGEN SYSTEMS	1,00	1,00
21	3871	22.	INSTRUMENTATION	0,00	1,00
22	3097	22.	INSTRUMENTATION	0,00	1,00
23	2636	22.	INSTRUMENTATION	1,00	1,00
24	2797	22.	INSTRUMENTATION	0,00	1,00
25	2789	22.	INSTRUMENTATION	0,00	1,00
26	3424	22.	INSTRUMENTATION	1,00	1,00
27	4022	22.	INSTRUMENTATION	0,00	1,00
28	2809	22.	INSTRUMENTATION	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	3094	22.	INSTRUMENTATION	0,00	1,00
30	2801	22.	INSTRUMENTATION	1,00	1,00
31	4026	22.	INSTRUMENTATION	0,00	1,00
32	3107	22.	INSTRUMENTATION	0,00	1,00
33	3090	22.	INSTRUMENTATION	0,00	1,00
34	2806	22.	INSTRUMENTATION	1,00	1,00
35	3085	22.	INSTRUMENTATION	0,00	1,00
36	17	22.2	MEASUREMENT OF AIR DATA PARAMETERS	1,00	1,00
37	1004	22.2.1	Pressure measurement	0,00	1,00
38	65	22.2.6	Airspeed indicator (ASI)	0,00	1,00
39	87	22.2.6	Airspeed indicator (ASI)	0,00	1,00
40	1690	22.2.6	Airspeed indicator (ASI)	0,00	1,00
			Total:42%	**17,00**	**40,00**

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: ATIALUK, Joseph

Location: Uganda Civil Aviation Authority

Your Result: 17,00 from 40,00 Points (42,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	12,00	20,00	60,00
21.2. AIRFRAME	3	3,00	3,00	100,00
21.3. HYDRAULICS	2	0,00	2,00	0,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	0,00	1,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	3	3,00	3,00	100,00
21.8. FUEL SYSTEM	2	1,00	2,00	50,00
21.9. ELECTRICS	2	1,00	2,00	50,00
21.13. OXYGEN SYSTEMS	2	2,00	2,00	100,00
22. INSTRUMENTATION	20	5,00	20,00	25,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	5	1,00	5,00	20,00
Total	**40**	**17,00**	**40,00**	**42,50**

1

Which statement is true regarding fouling of the spark plugs of an aircraft engine?

- ☐ Carbon fouling of the spark plugs is caused primarily by operating an engine at excessively high cylinder head temperatures.
- ☐ Spark plug fouling results from operating with an excessively rich mixture.
- ☒ Excessive heat in the combustion chamber of a cylinder causes oil to form on the center electrode of a spark plug and this fouls the plug.

ID:	2719(LTS:222939, LTM:1076)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	0,00

2

The purpose of a kill switch is to

- [ ] **shut off the fuel to the carburetor.**

- [ ] **ground the battery eliminating current for the ignition system.**

- [x] **ground the lead wire to the ignition coil shutting down the powerplant.**

ID:	3651(LTS:223016, LTM:1153)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	1,00

3

During which stroke of a reciprocating engine is the gaseous mixture expanding within the cylinder?

- ☐ **Power.**
- ☒ **Compression.**
- ☐ **Intake.**

ID:	2781(LTS:222961, LTM:1087)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	0,00

4

The formation of ice in a carburetor's throat is indicated by

☐ **a rapid increase in RPM, followed by rough engine operation.**

☒ **a drop in RPM, followed by rough engine operation.**

☐ **rough engine operation, followed by a decrease in oil pressure.**

ID:	3628(LTS:223015, LTM:1142)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	1,00

5

Many 4-cycle engines utilize what type of lubrication system?

- ☐ Forced.
- ☒ Fuel/oil mixture.
- ☐ Gravity.

ID:	3654(LTS:223018, LTM:1155)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	0,00

6

In a steep turn to the left, when using spoilers ...

- ☐ **The right aileron will ascend, the left one will descend, the right spoiler will retract and the left one will extend.**
- ☐ **The right aileron will ascend, the left one will descend, the right spoiler will extend and the left one will retract.**
- ☐ **The right aileron will descend, the left one will ascend, the right spoiler will extend and the left one will retract.**
- ☒ **The right aileron will descend, the left one will ascend, the right spoiler will retract and the left one will extend.**

ID:	1577(LTS:222614, LTM:371)	Max. Points	1,00
Syllabus reference:	21.2.3 Aeroplane: Wings, tail and control surfaces	Points	1,00

7

Among the different types of aircraft structures, the shell structures efficiently transmit the: 1. normal bending stresses 2. tangent bending stresses 3. torsional moment 4. shear stresses The combination regrouping all the correct statements is :

- ☒ **1, 2, 3**
- ☐ **1, 2, 4**
- ☐ **1, 3, 4**
- ☐ **2, 3, 4**

ID:	597(LTS:222375, LTM:131)	Max. Points	1,00
Syllabus reference:	21.2.3.1 Design and construction	Points	1,00

8

Upon which factor does wing loading during a level coordinated turn in smooth air depend?

- [x] **Angle of bank.**
- [ ] **Rate of turn.**
- [ ] **True airspeed.**

ID:	2422(LTS:222864, LTM:1001)	Max. Points	1,00
Syllabus reference:	21.2.3.3 Loads, stresses flutter	Points	1,00

9

The aircraft hydraulic system is designed to produce:

- ☐ **small pressure and large flow.**
- ☐ **high pressure and large flow.**
- ☐ **small pressure and small flow.**
- ☒ **high pressure and small flow.**

ID: 14(LTS:222246, LTM:2)
Syllabus reference: 21.3.2 Hydraulic systems

Max. Points 1,00
Points 0,00

10

The hydraulic oil, entering the hydraulic pump, is slightly pressurised to :

- ☒ **ensure sufficient pump output**
- ☐ **prevent overheating of the pump.**
- ☐ **prevent cavitation in the pump**
- ☐ **prevent vapour locking.**

ID:	900(LTS:222437, LTM:193)	Max. Points	1,00
Syllabus reference:	21.3.2.2 System components: etc	Points	0,00

11

A tubeless tyre is a tyre: 1. which requires solid or branched wheels 2. whose valve can be sheared in sudden accelerations 3. whose mounting rim must be flawless 4. which requires no rim protection between rim flange and tire removing device 5. which does not burst in the event of a tire puncture 6. which eliminates internal friction between the tube and the tire The combination regrouping all the correct statements is :

- [x] **3, 4, 5.**
- [ ] **2, 3, 6.**
- [ ] **1, 2, 5.**
- [ ] **1, 5, 6.**

ID:	1323(LTS:222546, LTM:302)	Max. Points	1,00
Syllabus reference:	21.4 LANDING GEAR, WHEELS, TYRES, BRAKES	Points	0,00

12

Assuming cabin pressure decreases, the cabin rate of climb indicator should indicate :

- ☐ **a rate of descent dependent upon the cabin differential pressure.**
- ☐ **zero.**
- ☐ **a rate of descent of approximately 300 feet per minutes.**
- ☒ **a rate of climb.**

ID:	25(LTS:222249, LTM:6)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	1,00

13

If the cabin altitude rises (aircraft in level flight), the differential pressure:

- ☐ **may exceed the maximum permitted differential unless immediate preventative action is taken.**
- ☒ **decreases**
- ☐ **remains constant**
- ☐ **increases**

ID:	836(LTS:222422, LTM:179)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	1,00

14

The pneumatic system accumulator is useful :

- ☐ in emergency cases.
- ☐ to eliminate the fluid flow variations.
- ☐ to offset for the starting of some devices.
- ☒ to eliminate the fluid pressure variations.

ID:	854(LTS:222428, LTM:184)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	1,00

15

Why should special precautions be taken when filling propane bottles?

- ☐ Propane is under high pressure when transferred from storage tanks to propane bottles.
- ☒ During transfer, propane reaches a high temperature and can cause severe burns.
- ☐ Propane is very cold and may cause freeze burns.

ID:	2929(LTS:222987, LTM:1124)	Max. Points	1,00
Syllabus reference:	21.8 FUEL SYSTEM	Points	0,00

16

When the combustion gases pass through a turbine the :

- ☐ temperature increases.
- ☒ pressure drops.
- ☐ velocity decreases.
- ☐ pressure rises.

ID:	1408(LTS:222567, LTM:324)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	1,00

17

If a current is passed through a conductor,

- ☐ **the current will increase.**
- ☒ **the intensity of the magnetic field will decrease.**
- ☐ **a force will be exerted on the conductor.**
- ☐ **there will be no effect unless the conductor is moved.**

ID:	926(LTS:222443, LTM:199)	Max. Points	1,00
Syllabus reference:	21.9.1.6 Electromagnetism	Points	0,00

18

The reason for using inverters in an electrical system is

- ☐ **To avoid a short circuit.**
- ☐ **To change the DC voltage.**
- ☐ **To change AC into DC.**
- ☒ **To change DC into AC.**

ID:	817(LTS:222420, LTM:176)	Max. Points	1,00
Syllabus reference:	21.9.3.1 DC Generation	Points	1,00

19

If a public transport jet aeroplane is operated at FL 450, with 180 passenger seats, what will be the number of masks available for use

- ☐ **240 (one additional mask per seat block).**
- ☐ **210 (one additional mask per seat row).**
- ☐ **270 (150% of the seating capacity).**
- ☒ **198 (110% of the seating capacity).**

ID:	883(LTS:222432, LTM:188)	Max. Points	1,00
Syllabus reference:	21.13 OXYGEN SYSTEMS	Points	1,00

20

The state in which the breathing oxygen for the cockpit of jet transport aeroplanes is stored is :

- ☐ **Gaseous or chemical compound..**
- ☐ **Liquid.**
- ☐ **Chemical compound.**
- ☒ **Gaseous.**

ID:	1601(LTS:222619, LTM:376)	Max. Points	1,00
Syllabus reference:	21.13 OXYGEN SYSTEMS	Points	1,00

21

When an altimeter is changed from 30.11" Hg to 29.96" Hg, in which direction will the indicated altitude change and by what value?

- ☐ **Altimeter will indicate 150 feet lower.**
- ☒ **Altimeter will indicate 150 feet higher.**
- ☐ **Altimeter will indicate 15 feet lower.**

ID:	3871(LTS:223025, LTM:1162)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

22

How can a pilot identify a lighted heliport at night?

- ☐ Green, yellow, and white beacon light.
- ☐ Green and white beacon light with dual flash of the white.
- ☒ White and red beacon light with dual flash of the white.

ID:	3097(LTS:222993, LTM:1130)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

23

A gyroplane will have the greatest tendency to roll during

- ☐ **descending flight in which forward airspeed decreases.**
- ☐ **climbing flight in which forward airspeed decreases.**
- ☒ **horizontal flight at high speed.**

ID:	2636(LTS:222937, LTM:1074)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

24

Deviation error of the magnetic compass is caused by

- ☐ northerly turning error.
- ☒ the difference in location of true north and magnetic north.
- ☐ certain metals and electrical systems within the aircraft.

ID:	2797(LTS:222966, LTM:1092)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

25

In the Northern Hemisphere, a magnetic compass will normally indicate a turn toward the north if

- ☐ an aircraft is decelerated while on an east or west heading.
- ☒ a left turn is entered from a west heading.
- ☐ an aircraft is accelerated while on an east or west heading.

ID:	2789(LTS:222964, LTM:1090)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

26

Which procedure is recommended to prevent or overcome spatial disorientation?

- [x] **Rely entirely on the indications of the flight instruments.**
- [ ] **Reduce head and eye movements to the greatest extent possible.**
- [ ] **Avoid steep turns and rough control movements.**

ID:	3424(LTS:223010, LTM:1137)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

See Attachments:		
1. instrument_82.jpg	figure 82	(Attachment No.1)

27

(Refer to Figure 82.) Which is an acceptable range of accuracy when performing an operational check of dual VOR's using one system against the other?

- [x] **1**
- [ ] **4**
- [ ] **2**

ID:	4022(LTS:223026, LTM:1163)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

28

Which instrument would be affected by excessively low pressure in the airplane's vacuum system?

- ☐ **Airspeed indicator.**
- ☒ **Pressure altimeter.**
- ☐ **Heading indicator.**

ID:	2809(LTS:222970, LTM:1096)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

29

How can a pilot identify a military airport at night?

- ☐ **Green and white beacon light with dual flash of the white.**
- ☒ **White and red beacon light with dual flash of the white.**
- ☐ **Green, yellow, and white beacon light.**

ID:	3094(LTS:222992, LTM:1129)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

30

What should be the indication on the magnetic compass as you roll into a standard rate turn to the right from a south heading in the Northern Hemisphere?

☐ **The compass will initially indicate a turn to the left.**

☐ **The compass will remain on south for a short time, then gradually catch up to the magnetic heading of the airplane.**

☒ **The compass will indicate a turn to the right, but at a faster rate than is actually occurring.**

ID:	2801(LTS:222968, LTM:1094)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

31

Which is the maximum tolerance for the VOR indication when the CDI is centered and the aircraft is directly over the airborne checkpoint?

☐ **Plus or minus 6 degrees of the designated radial.**

☒ **Plus or minus 4 degrees of the designated radial.**

☐ **Plus 6degrees or minus 4 degrees of the designated radial.**

ID:	4026(LTS:223028, LTM:1165)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

32

What are the indications of the pulsating VASI?

☒ **High - pulsing white, on glidepath - green, low - pulsing red.**

☐ **High - pulsing white, on course and on glidepath - steady white, off course but on glidepath - pulsing white and red; low - pulsing red.**

☐ **High - pulsing white, on glidepath - steady white, slightly below glide slope steady red, low - pulsing red.**

ID:	3107(LTS:223006, LTM:1133)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

33

Identify taxi leadoff lights associated with the centerline lighting system.

- ☒ **Alternate green and yellow lights curving from the centerline of the runway to the centerline of the taxiway.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to a point on the exit.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to the edge of the taxiway.**

ID:	3090(LTS:222991, LTM:1128)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

34

Which statement is true about magnetic deviation of a compass?

- ☒ **Deviation varies for different headings of the same aircraft.**
- ☐ **Deviation is the same for all aircraft in the same locality.**
- ☐ **Deviation is different in a given aircraft in different localities.**

ID:	2806(LTS:222969, LTM:1095)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

35

Identify touchdown zone lighting (TDZL).

☐ **Flush centerline lights spaced at 50-foot intervals extending through the touchdown zone.**

☐ **Two rows of transverse light bars disposed symmetrically about the runway centerline.**

☒ **Alternate white and green centerline lights extending from 75 feet from the threshold through the touchdown zone.**

ID:	3085(LTS:222989, LTM:1126)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

36

If the Pitot Head and Static Vent were blocked by ice, which instruments would be affected?

- ☐ **ASI, Altimeter and Slip indicator.**
- ☐ **The ASI would under read.**
- ☒ **Altimeter, VSI and ASI would give inaccurate readings.**

ID:	17(LTS:222248, LTM:4)	Max. Points	1,00
Syllabus reference:	22.2 MEASUREMENT OF AIR DATA PARAMETERS	Points	1,00

37

A simple pressure altimeter uses:

- ☐ pitot pressure fed to an open capsule
- ☐ static pressure fed outside a completely evacuated aneroid capsule.
- ☒ static pressure fed to an open capsule
- ☐ static pressure fed into a gas tight casing, but outside a partially evacuated capsule

ID:	1004(LTS:222473, LTM:220)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

38

If the Pitot opening is blocked, which instruments would be affected (separate static vent)?

- ☐ **ASI and VSI.**
- ☒ **ASI, Altimeter and VSI.**
- ☐ **ASI only.**

ID:	65(LTS:222259, LTM:16)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

39

Rectified Airspeed is:

- ☐ **IAS corrected for instrument and pressure error.**
- ☒ **IAS corrected for density and compressibility errors.**
- ☐ **IAS corrected for density error.**

ID:	87(LTS:222263, LTM:20)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

40

VFE is the maximum speed:

- ☐ *at which the flaps can be operated when flying in turbulences.*
- ☐ *with the flaps extended.*
- ☐ *with the flaps extended in the take-off configuration.*
- ☒ *with the flaps extended in the landing configuration.*

ID:	1690(LTS:222657, LTM:404)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

Attachment No. 1

Questions 27
figure 82
instrument_82.jpg

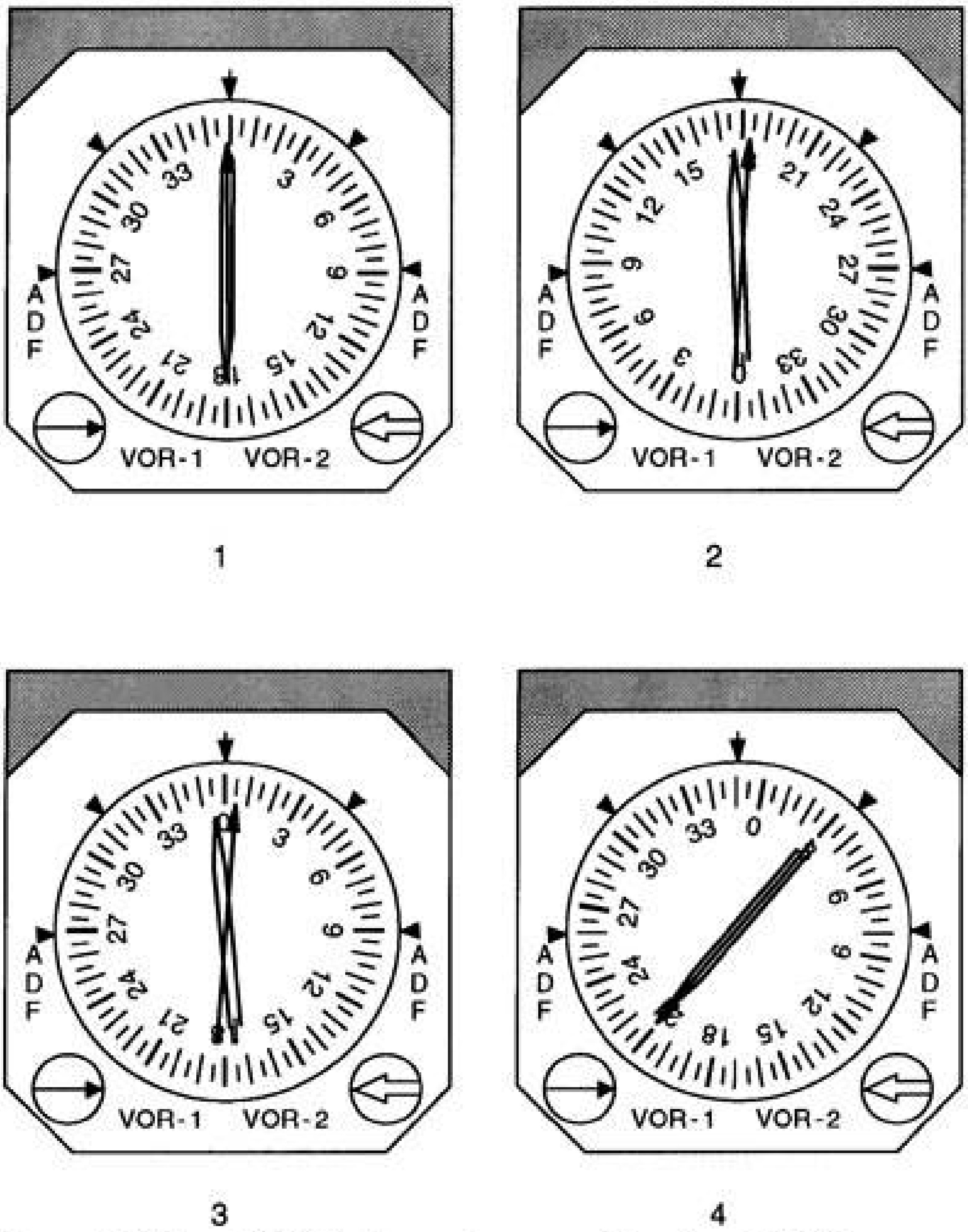

Figure 82. Dual VOR System, Accuracy Check © ASA